Technical Project Log: Cessna 210 Airframe Reconstruction

NASA Variant Parametric Modeling & Aerodynamic Analysis

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1 Project Introduction

This log documents the 3D reconstruction and numerical discretization of a Cessna 210 Centurion airframe using NASA's OpenVSP (Vehicle Sketch Pad). The primary goal was to translate 2D engineering blueprints into a watertight 3D parametric model suitable for future avionics and telemetry integration within the Team Antarez framework.

2 Geometric Modeling & Parameter Verification

The airframe was constructed using a multi-component assembly. A significant portion of the workflow involved the cross-verification of provided instructional dimensions against original 3-view drawings.

Component	**Finalized Dimension**	**Status/Notes**
Main Wingspan	36.748 ft	Verified (NACA 2412)
Fuselage Length	24.580 ft	Scaled to Reference
Main Wing Root Chord	5.525 ft	Parameterized
Horizontal Tail Span	12.990 ft	Verified
Vertical Tail Height	**7.10 ft**	**Corrected from 2.94 ft typo**

Table 1: Finalized Design Parameters for Cessna 210 Model.

2.1 Correction of Specification Errors

During the construction of the empennage, a major discrepancy was identified in the source instructions, which listed the vertical stabilizer height at 2.94 ft. Through visual ratio analysis of the NASA blueprints, this was identified as a scaling typo and corrected to ∼7.10 ft to maintain aerodynamic stability and geometric fidelity.

3 Numerical Mesh Generation

The model was discretized into a Vortex Lattice Method (VLM) mesh to analyze lift-induced pressure gradients.

- **Discretization:** A total of 5,731 nodes were generated to ensure surface continuity.
- **Wing Geometry:** Implemented 2.0° dihedral for lateral stability.
- **Solver Configuration:** Performed an Alpha sweep from 0° to 10° ($N_{pts} = 11$) to characterize the lift-slope derivative and drag polar.

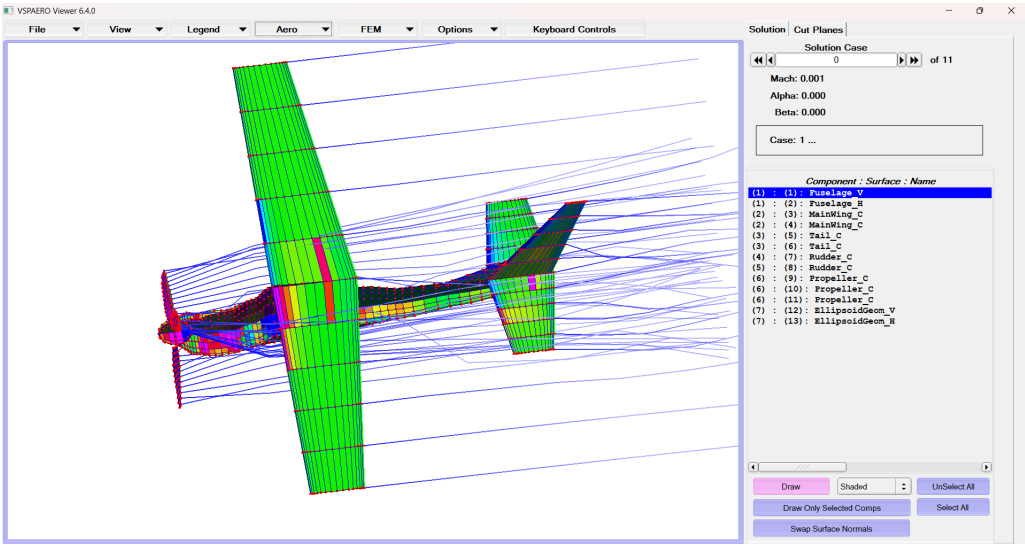

Figure 1: Converged VSPAERO solution showing pressure coefficient (C_p) distribution and wake streamlines.

4 Computational Analysis & Results

The simulation achieved numerical convergence by optimizing trailing-edge discretization. Initial Kutta condition detection failures ($N_k = 0$) were resolved by implementing a "Skew Both" closure at the trailing edges, allowing the solver to successfully identify 109 Kutta nodes.

- **Pressure Mapping:** Surface C_p analysis confirms appropriate lift generation across the NACA 2412 sections.
- **Asymmetric Loading Analysis:** Localized pressure variations on the wing surfaces were identified as asymmetric flow impingement from the 3-blade propeller wake. The solver accurately captured the interference between the discrete prop-geometry and the wing panels.
- **Solver Stability:** The model demonstrated high stability, with the log error for CL_{tot} decreasing significantly across iterations, verifying a physically consistent solution.
- **Wake Visualization:** Enabled wake streamlines to verify the downwash effect, confirming the lifting-line theory application.

5 Conclusion

The project successfully transitioned from geometric reconstruction to a verified aerodynamic dataset. By debugging the numerical instabilities and managing junction interference, a simulation-ready digital twin has been established. This model will serve as the baseline for Team Antarez hardware-in-the-loop (HIL) testing and flight controller calibration.